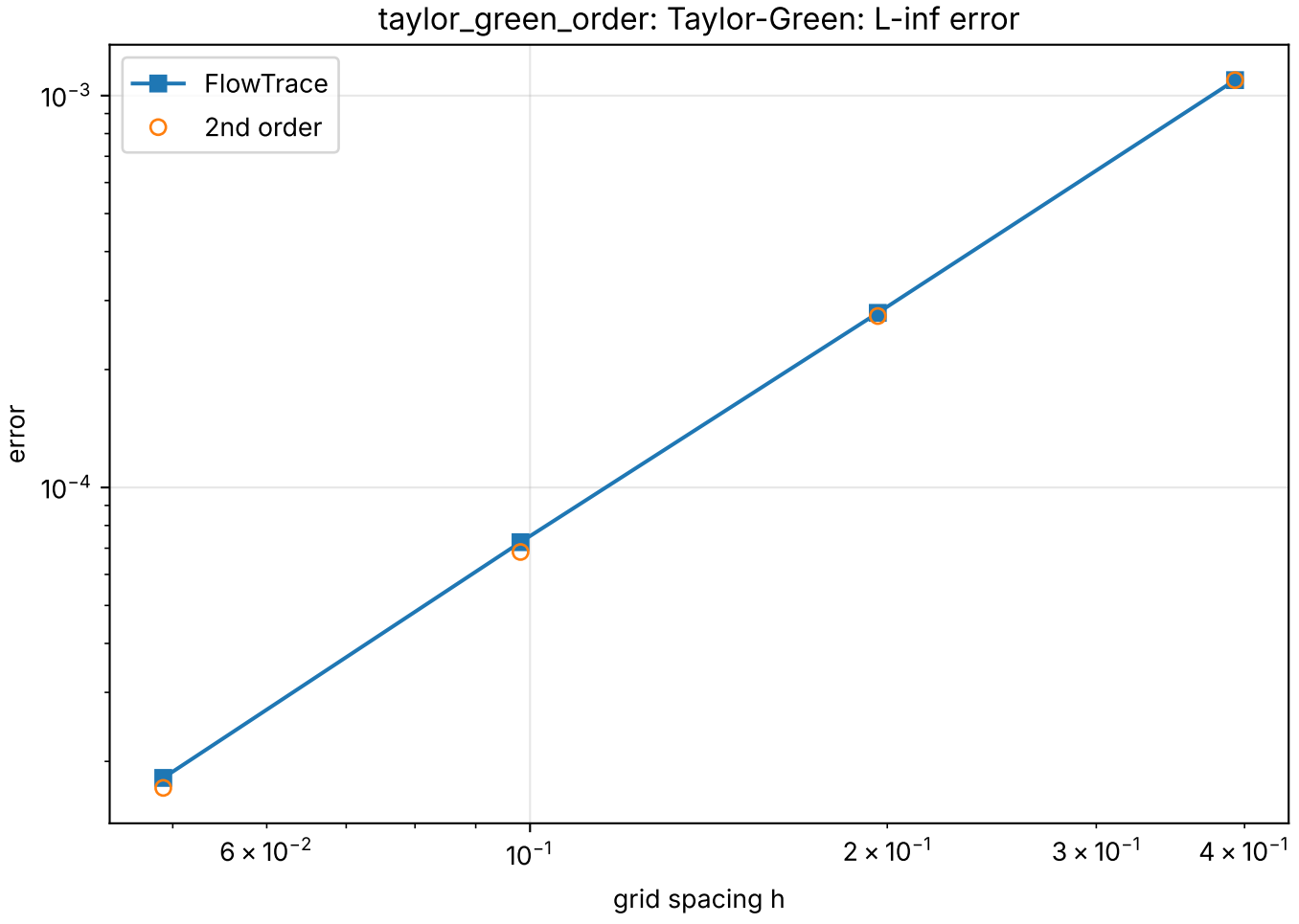
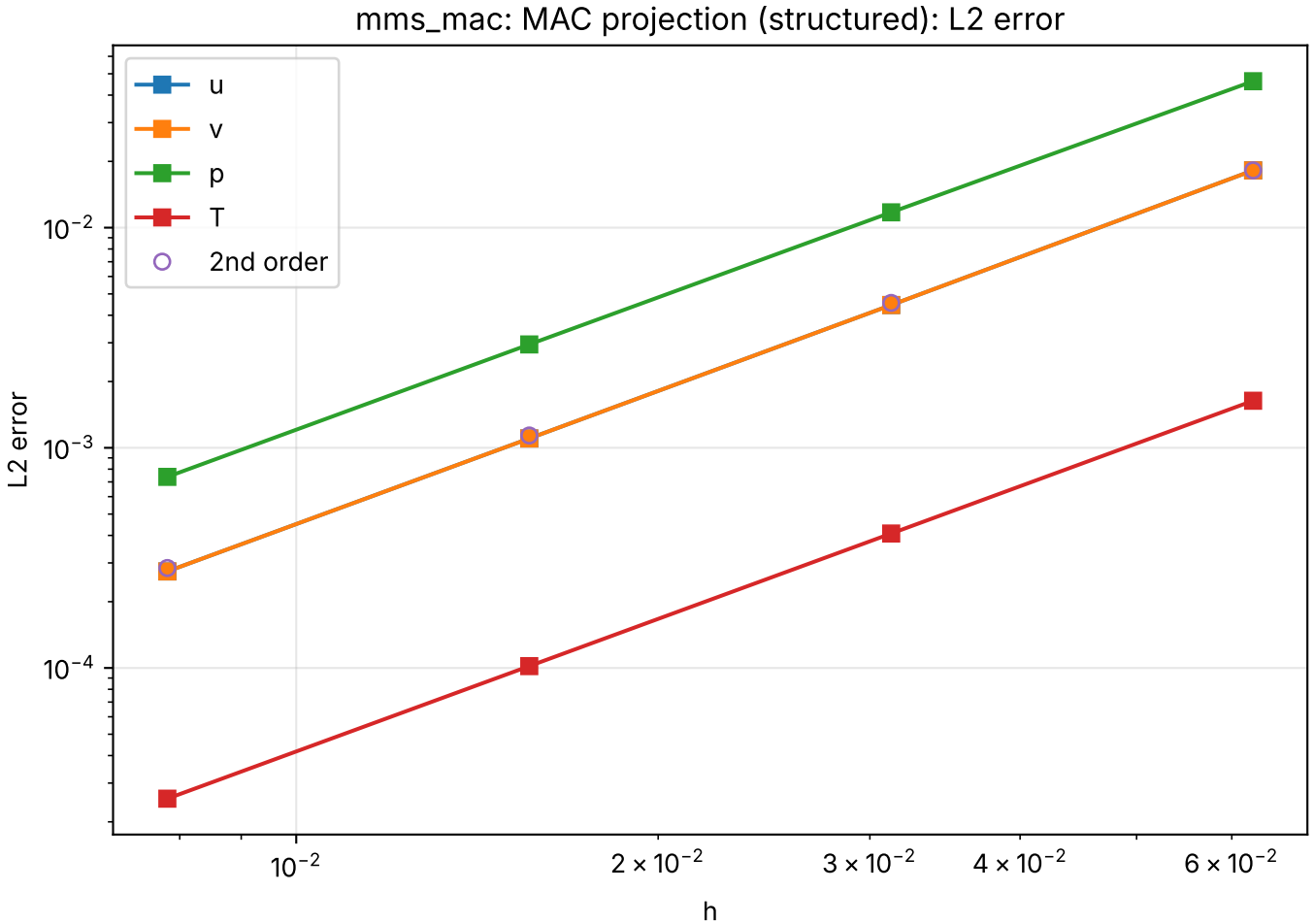
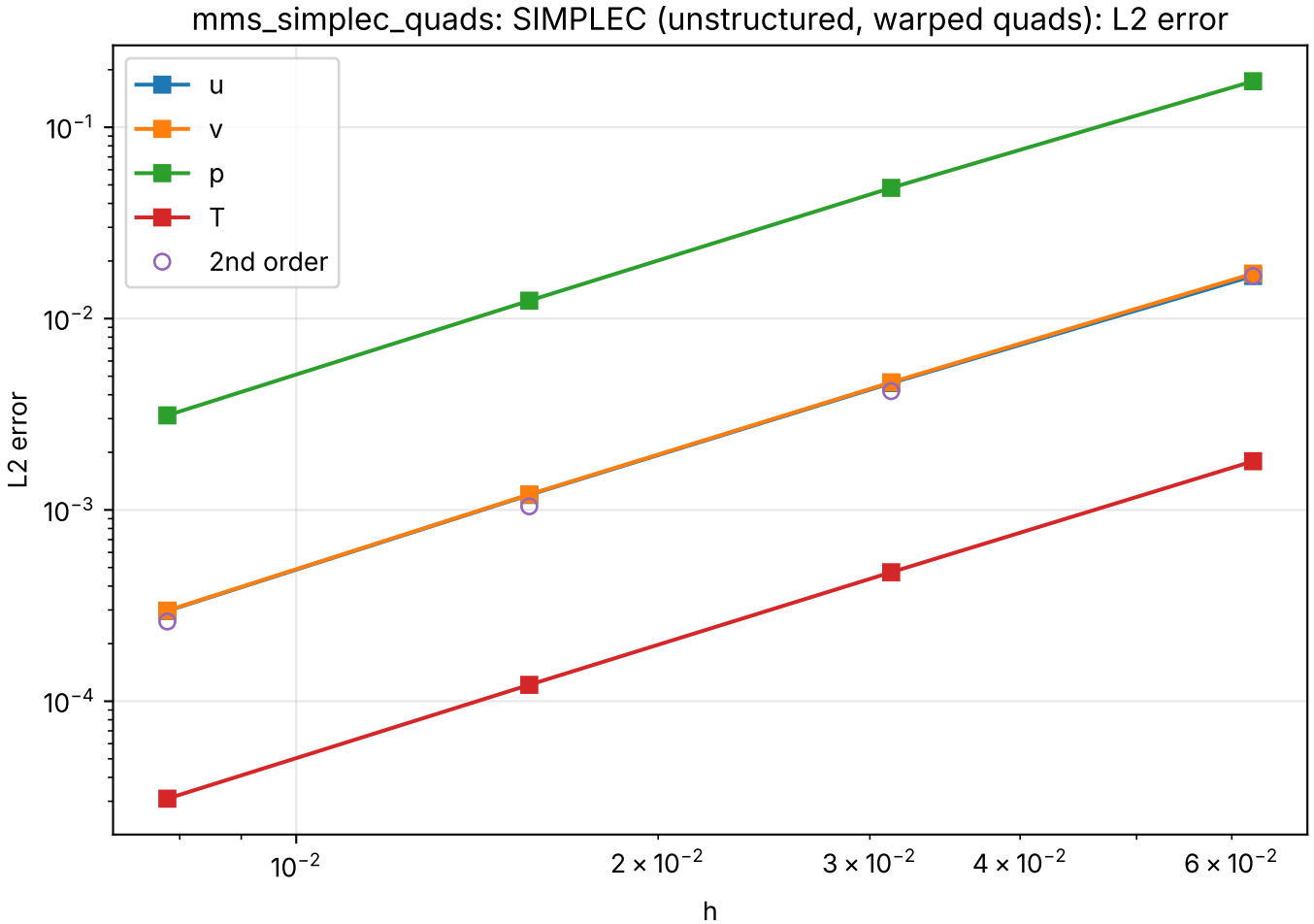
```
FlowTrace validation report  (full tier)

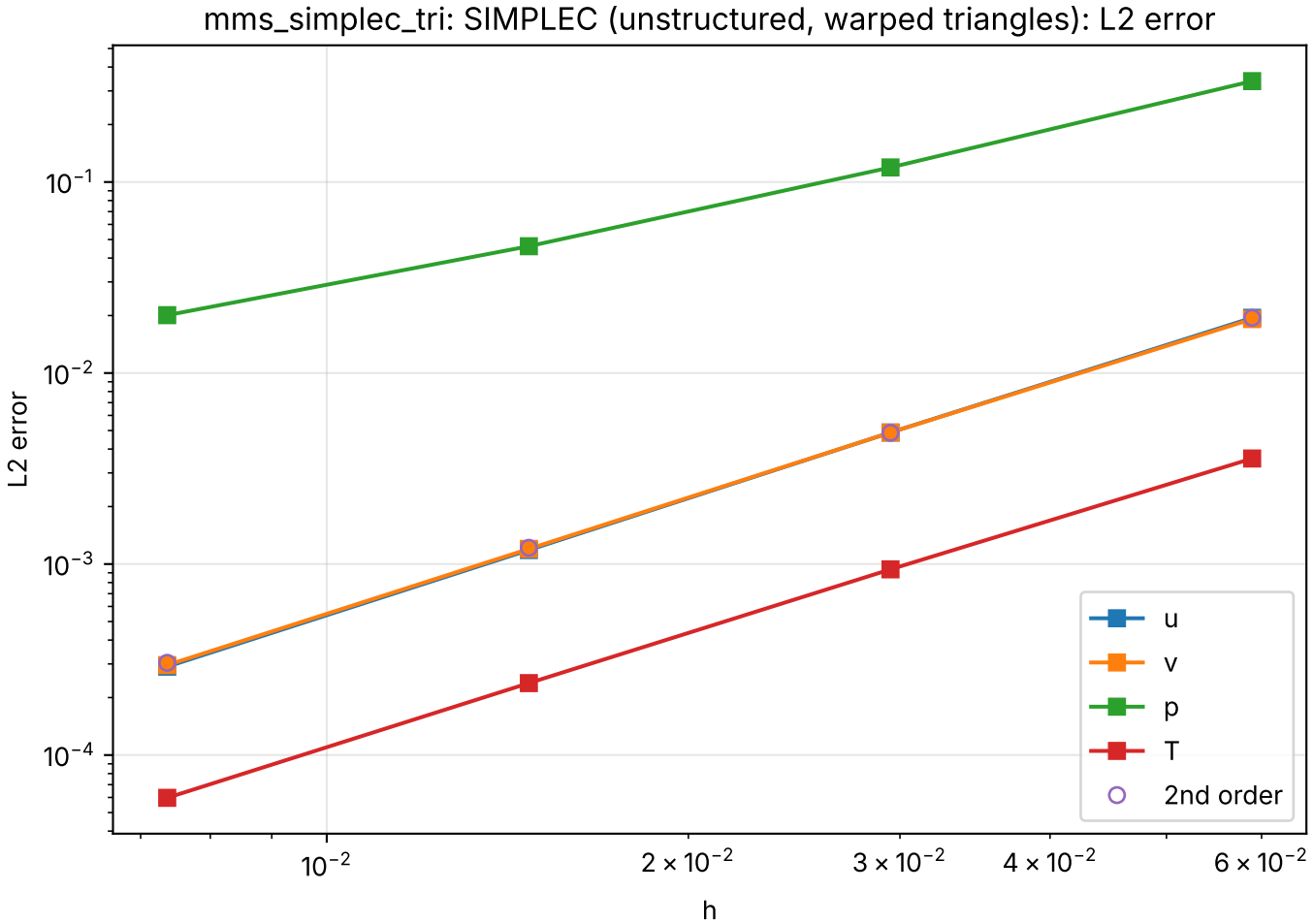
2026-10-27T06:45:34  FlowTrace 0.3.0
Intel(R) Xeon(R) Processor @ 2.80GHz (4C/4T)
GPU: none used

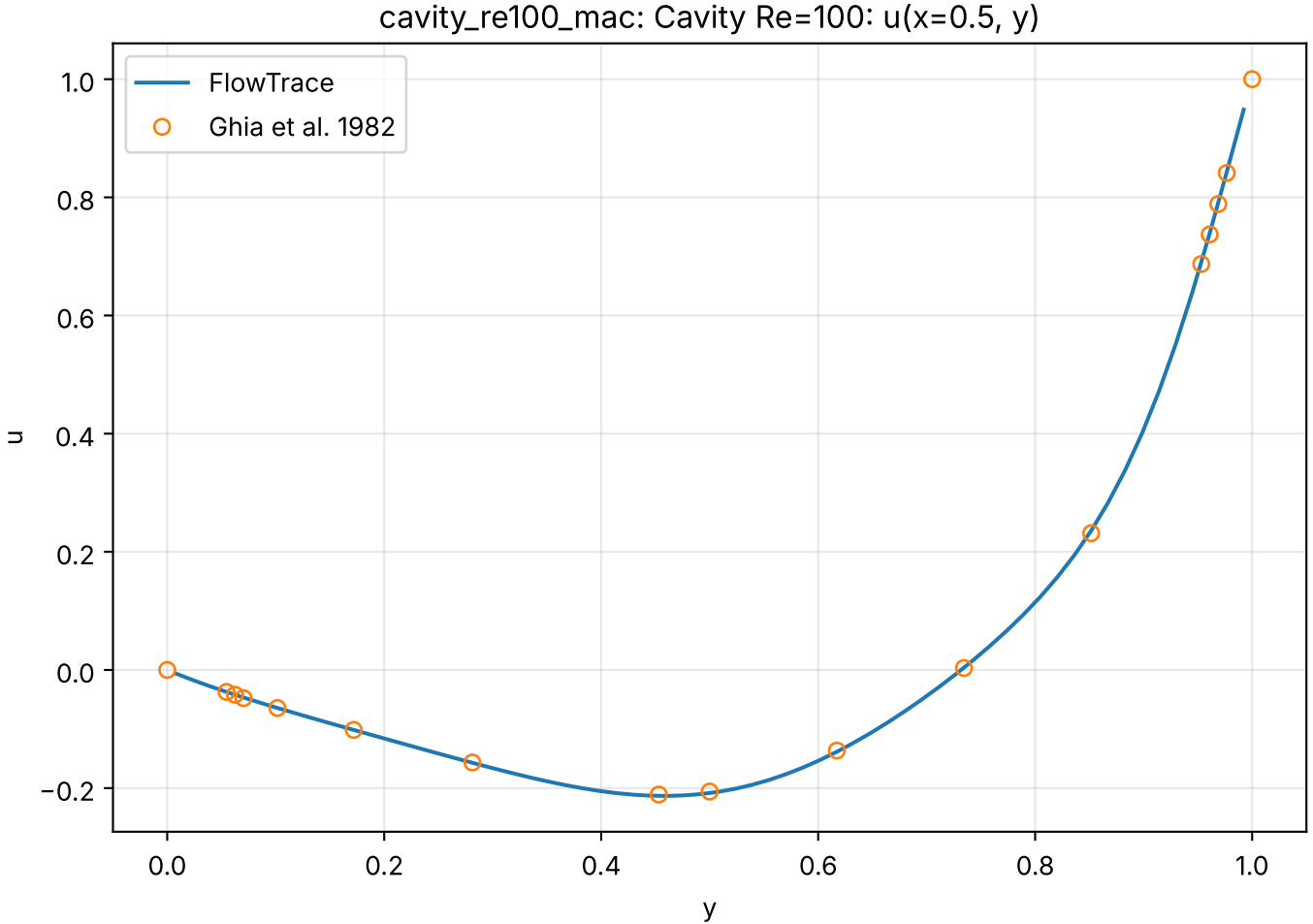
[PASS] Taylor-Green vortex: error vs exact solution
      Linf error n=16: 0.001095 |u-u_exact|
      Linf error n=32: 2.790e-04 |u-u_exact|
      Linf error n=64: 7.252e-05 |u-u_exact|
      Linf error n=128: 1.815e-05 |u-u_exact|
      observed order (finest pair): 1.998   (ref 2)
[PASS] MMS: MAC solver observed order (walls, energy)
      observed order u (L2): 2.006   (ref 2)
      observed order v (L2): 2.006   (ref 2)
      observed order p (L2): 1.998   (ref 2)
      observed order T (L2): 2   (ref 2)
      L2 error u, finest grid: 2.744e-04
      L2 error v, finest grid: 2.744e-04
      L2 error p, finest grid: 7.379e-04
      L2 error T, finest grid: 2.548e-05
[PASS] MMS: SIMPLEC observed order on warped (non-orthogonal) quads
      observed order u (L2): 2.015   (ref 2)
      observed order v (L2): 2.016   (ref 2)
      observed order p (L2): 1.991   (ref 2)
      observed order T (L2): 1.977   (ref 2)
      L2 error u, finest grid: 2.964e-04
      L2 error v, finest grid: 2.978e-04
      L2 error p, finest grid: 0.00312
      L2 error T, finest grid: 3.095e-05
[PASS] MMS: SIMPLEC observed order on warped triangles
      observed order u (L2): 2.028   (ref 2)
      observed order v (L2): 2.021   (ref 2)
      observed order p (L2): 1.199   (ref 2)
      observed order T (L2): 1.995   (ref 2)
      L2 error u, finest grid: 2.903e-04
      L2 error v, finest grid: 2.959e-04
      L2 error p, finest grid: 0.02009
      L2 error T, finest grid: 5.972e-05
[PASS] Discrete continuity of the projection
      max|div u| fast separable solver: 6.828e-15 1/s  (ref 0)
      max|div u| multigrid-PCG (obstacle): 3.400e-12 1/s  (ref 0)
[PASS] Lid-driven cavity Re=100, MAC 64x64
      max |du| centreline: 0.004195   (ref 0)
      max |dv| centreline: 0.008457   (ref 0)
[PASS] Lid-driven cavity Re=100, SIMPLEC 48x48 quads
      max |du| centreline: 0.004236   (ref 0)
      max |dv| centreline: 0.008108   (ref 0)
[PASS] Lid-driven cavity Re=100, SIMPLEC 48x48 skewed triangles
      max |du| centreline: 0.005587   (ref 0)
      max |dv| centreline: 0.008918   (ref 0)
      max non-orthogonality: 59.55 deg
[PASS] Lid-driven cavity Re=1000, SIMPLEC 128x128 quads
      max |du| centreline: 0.005344   (ref 0)
      max |dv| centreline: 0.009979   (ref 0)
      iterations to 1e-6: 3993
      wall time: 186 s
[PASS] Natural convection Ra=1e4, MAC 64x64
      Nu (hot wall): 2.251   (ref 2.243)
      u_max on x=0.5: 16.19   (ref 16.18)
      v_max on y=0.5: 19.64   (ref 19.62)
[PASS] Natural convection Ra=1e5, MAC 64x64 with FMG
      Nu (hot wall): 4.569   (ref 4.519)
[PASS] Natural convection Ra=1e3, SIMPLEC + Boussinesq 48x48
      Nu (hot wall): 1.118   (ref 1.118)
      energy imbalance |sum Q| / Q_hot: 3.496e-13   (ref 0)
[PASS] Channel flow Re=50: Poiseuille profile and pressure gradient
      outlet profile max |u - u_Poiseuille|: 0.005777   (ref 0)
      dp/dx (developed region): -0.2383   (ref -0.24)
[PASS] SIMPLEC global conservation (channel, exported step mesh)
      net mass imbalance / inflow, channel: 8.318e-11   (ref 0)
      net mass imbalance / inflow, step mesh: 2.878e-10   (ref 0)
[PASS] Grid convergence index: Nu, natural convection Ra=1e4
      Nu n=32: 2.272   (ref 2.243)
      Nu n=64: 2.251   (ref 2.243)
      Nu n=128: 2.246   (ref 2.243)
      observed order p: 2.137   (ref 2)
      Richardson-extrapolated Nu: 2.245   (ref 2.243)
      GCI fine grid: 0.0783 %
      asymptotic-range ratio: 1   (ref 1)
[PASS] Grid convergence index: u_min on the cavity centreline, Re=100
      u_min n=32: -0.2104   (ref -0.2109)
      u_min n=64: -0.2132   (ref -0.2109)
      u_min n=128: -0.2138   (ref -0.2109)
      observed order p: 2.106   (ref 2)
      Richardson-extrapolated u_min: -0.214
      GCI fine grid: 0.1136 %
[PASS] Turbulent channel Re_tau=392, k-omega SST (y+ ~ 0.25) vs DNS
      k-omega-sst: max |U+ - DNS| (y+ >= 1): 4.066 %  (ref 0)
      k-omega-sst: centreline U+ error: -1.718 %  (ref 0)
      k-omega-sst: wall shear (exact 1 at equilibrium): 0.9994   (ref 1)
      k-omega-sst: iterations: 3257
[PASS] Turbulent channel Re_tau=392, k-epsilon + wall functions vs DNS
      k-epsilon-standard: max |U+ - DNS| (y+ >= 51): 1.874 %  (ref 0)
      k-epsilon-standard: centreline U+ error: 0.2964 %  (ref 0)
      k-epsilon-standard: wall shear (exact 1 at equilibrium): 0.9999   (ref 1)
      k-epsilon-standard: iterations: 4992
      k-epsilon-realizable: max |U+ - DNS| (y+ >= 51): 5.839 %  (ref 0)
      k-epsilon-realizable: centreline U+ error: 5.5 %  (ref 0)
      k-epsilon-realizable: wall shear (exact 1 at equilibrium): 0.9999   (ref 1)
      k-epsilon-realizable: iterations: 5764
[PASS] Turbulent flat plate Re_L=5e6 (NASA TMR set-up), k-omega SST
      Cf at x = 0.97 (Re_x = 4.85e6): 0.002614   (ref 0.002875)
      mean deviation from White, 1e6 < Re_x < 1e7: -9.844 %  (ref 0)
      mean deviation from Schultz-Grunow, same range: -5.317 %  (ref 0)
```

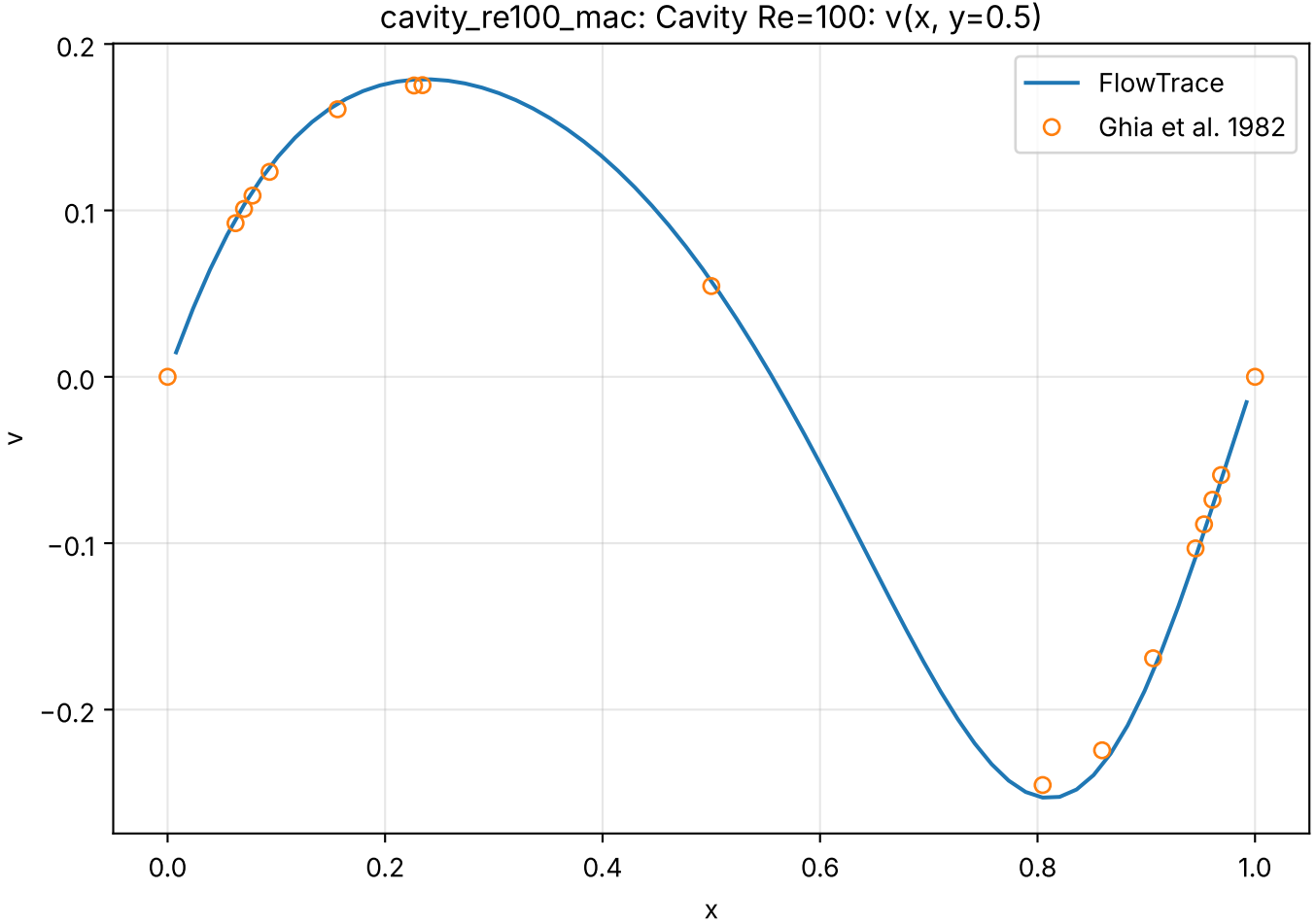

taylor_green_order: Taylor-Green: L-inf error
FlowTrace
2nd order
10^{-3}
10^{-4}
error
6×10^{-2}
10^{-1}
2×10^{-1}
3×10^{-1}
4×10^{-1}
grid spacing h

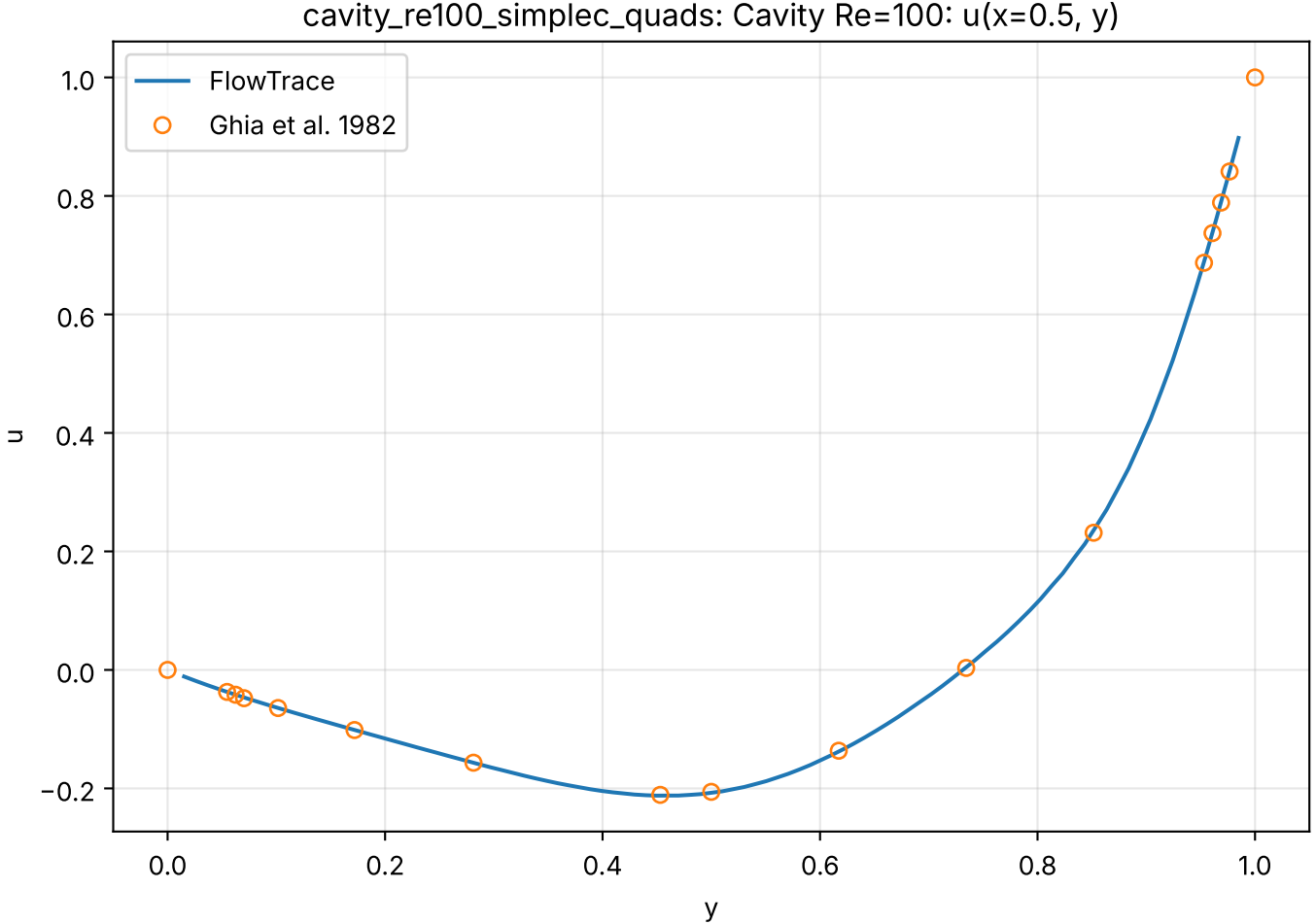

mms_mac: MAC projection (structured): L2 error
u
v
p
T
2nd order
L2 error
10^{-2}
10^{-3}
10^{-4}
10^{-2}
2×10^{-2}
3×10^{-2}
4×10^{-2}
6×10^{-2}
h

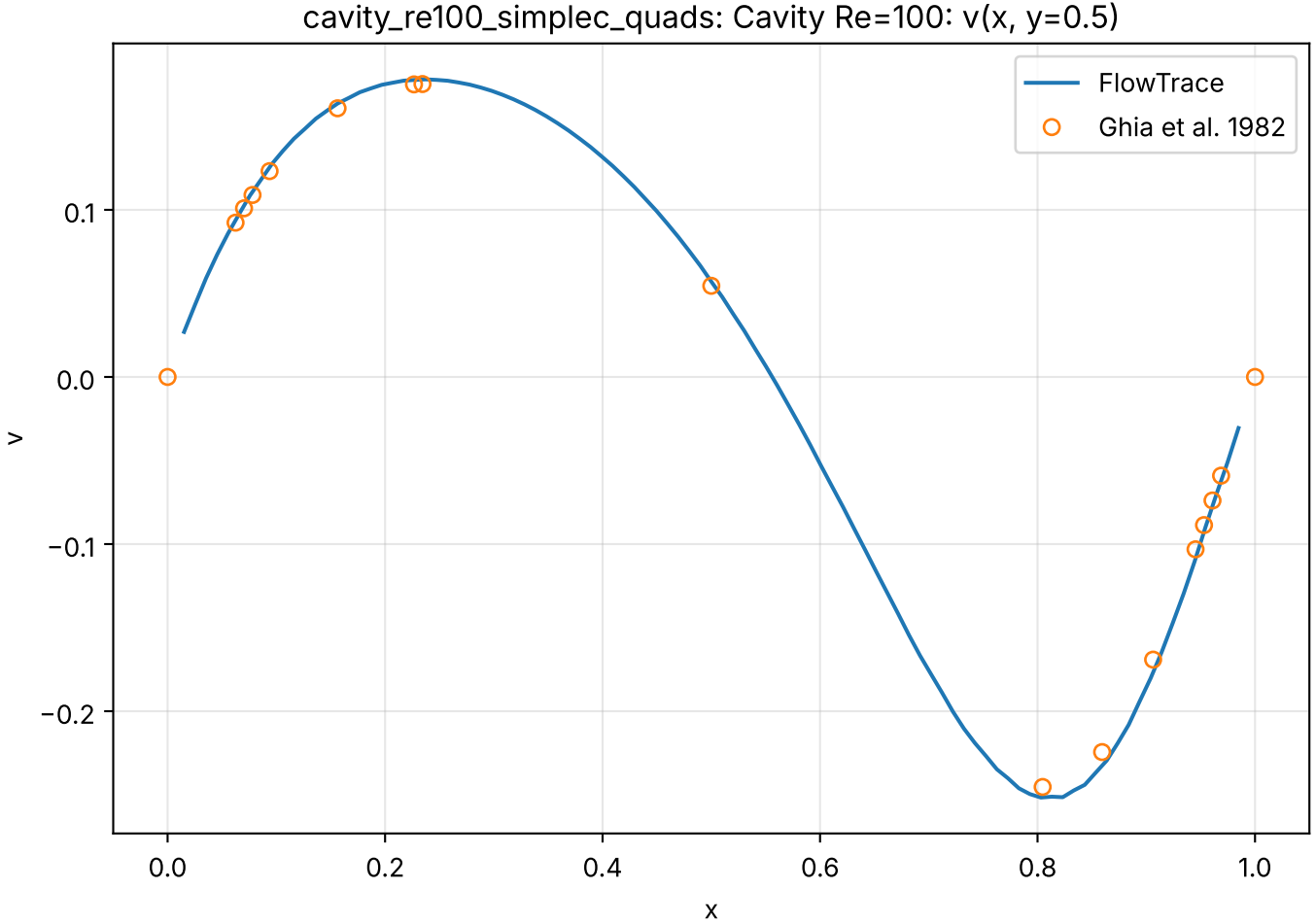

mms_simplec_quads: SIMPLEC (unstructured, warped quads): L2 error
u
v
p
T
2nd order
L2 error
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-2}
2×10^{-2}
3×10^{-2}
4×10^{-2}
6×10^{-2}
h

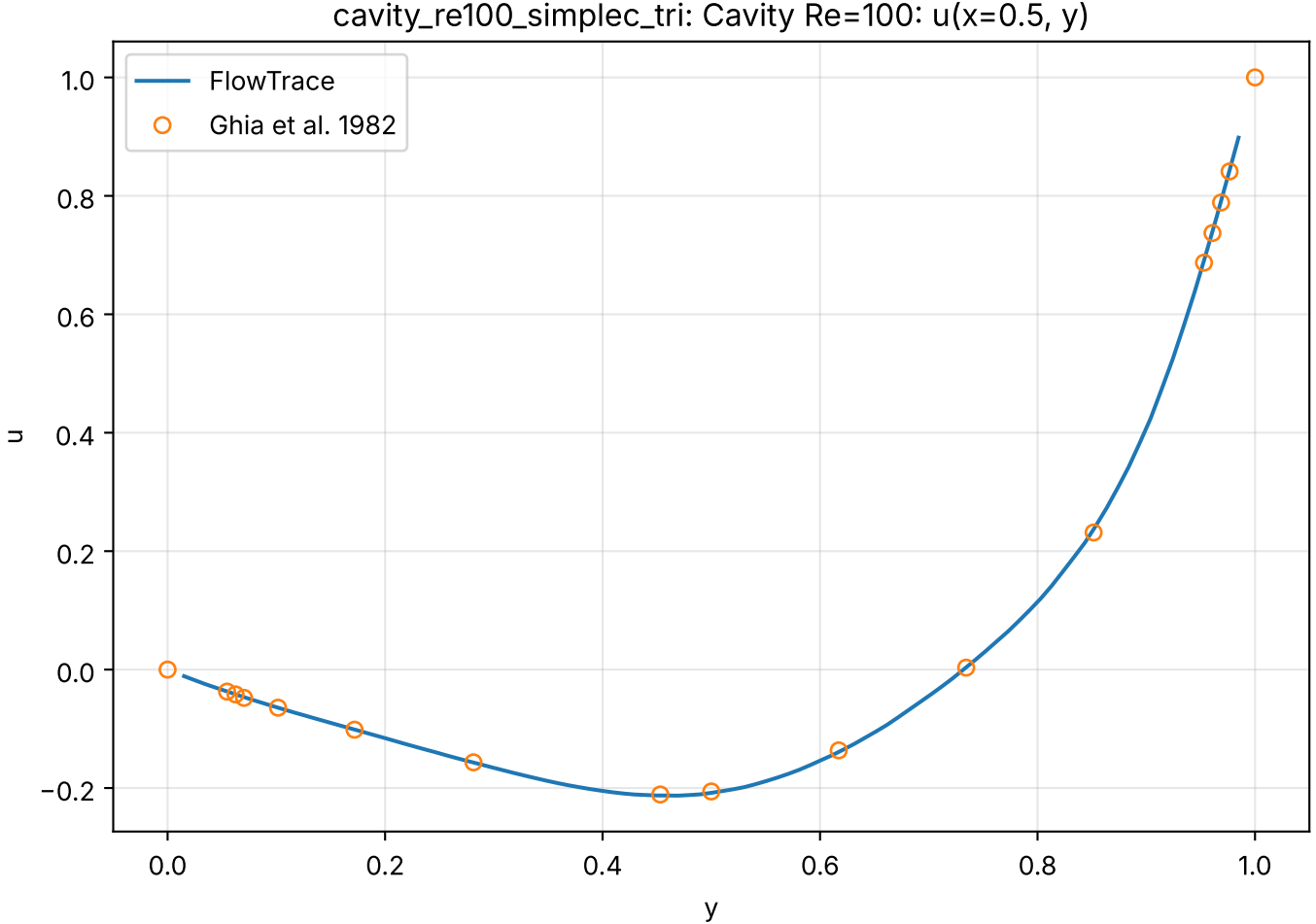

mms_simplec_tri: SIMPLEC (unstructured, warped triangles): L2 error
L2 error
h
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-2}
2×10^{-2}
3×10^{-2}
4×10^{-2}
6×10^{-2}
u
v
p
T
2nd order

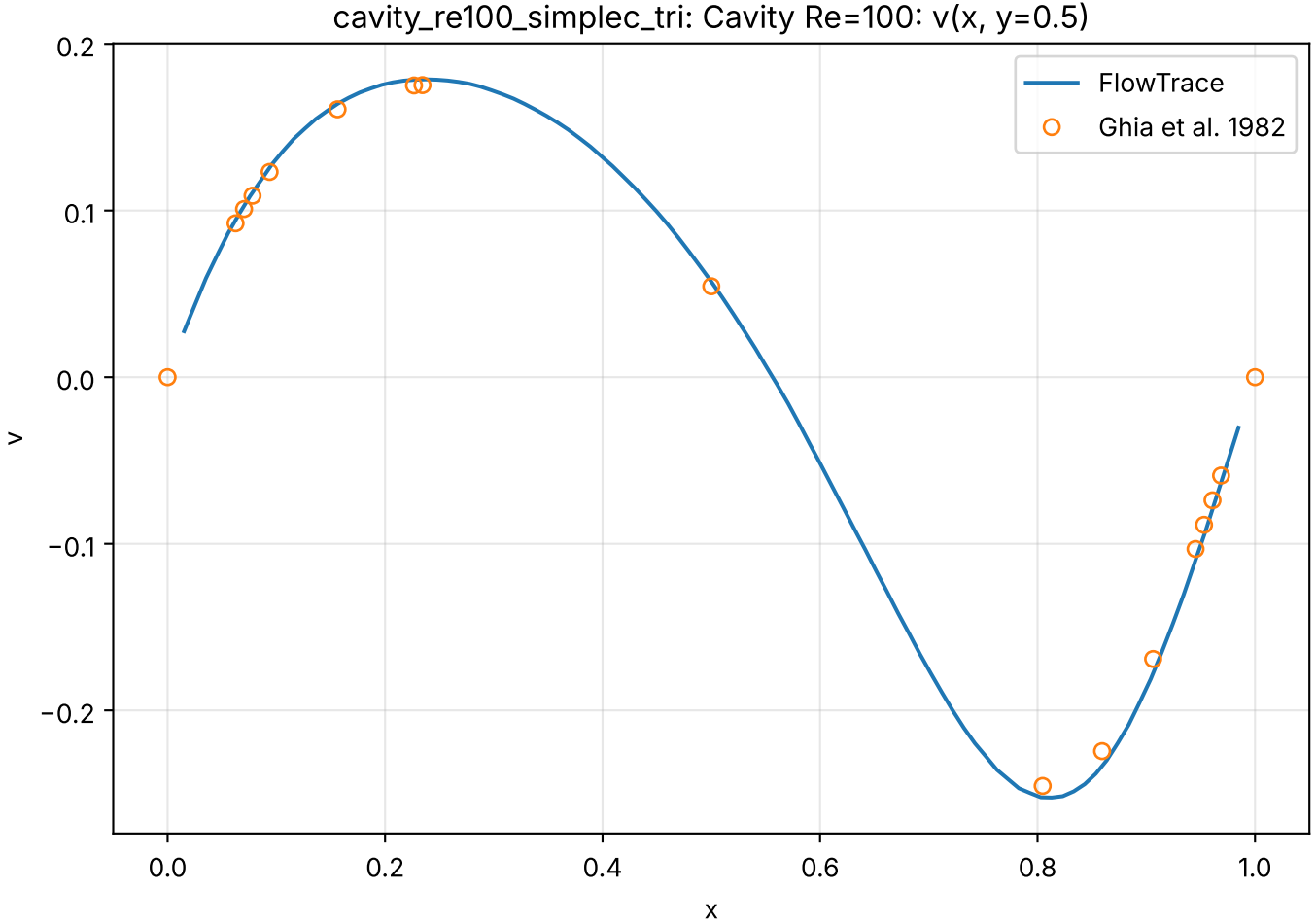

cavity_re100_mac: Cavity Re=100: u(x=0.5, y)
FlowTrace
Ghia et al. 1982
u
y
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
0.0
0.2
0.4
0.6
0.8
1.0

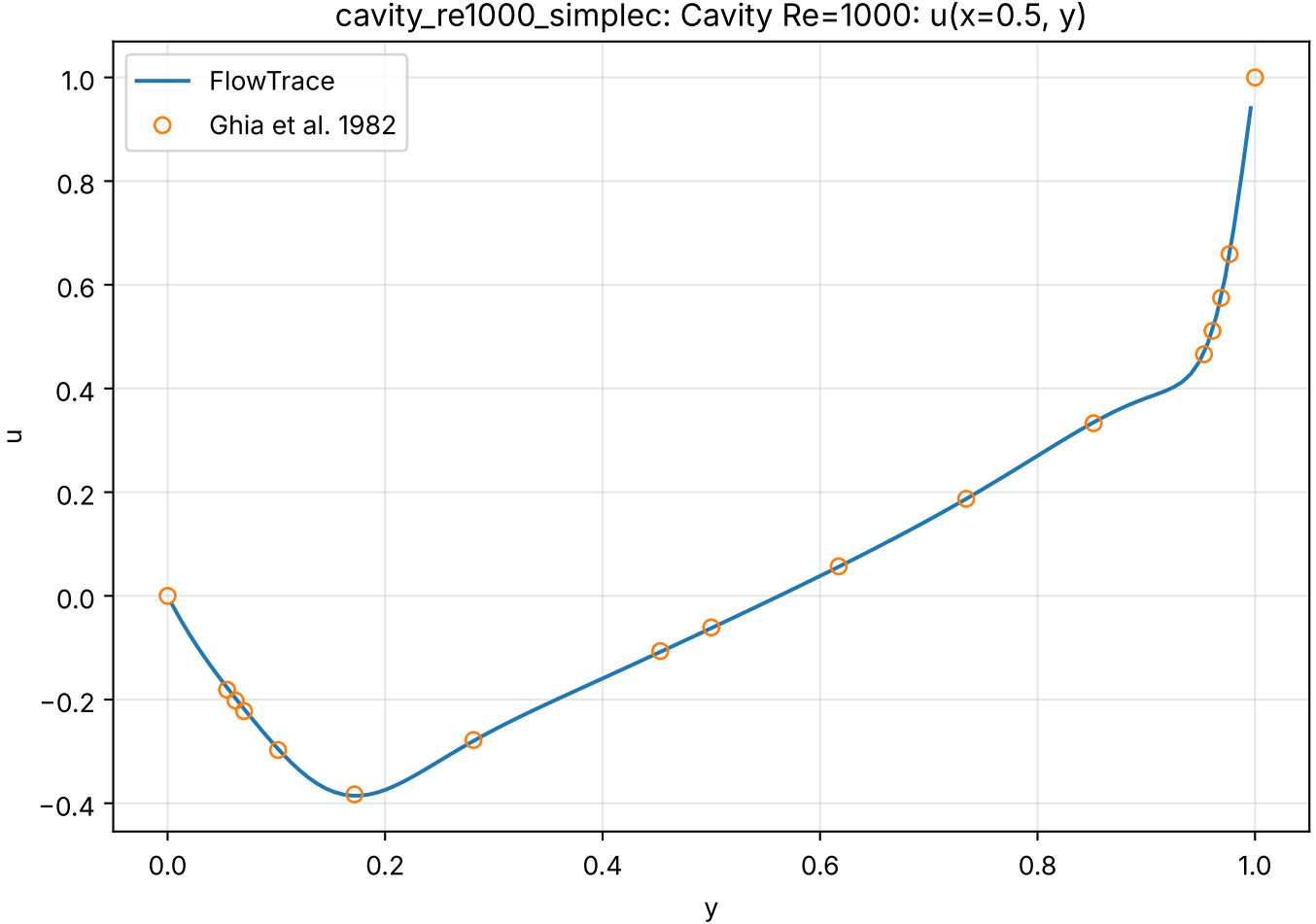

cavity_re100_mac: Cavity Re=100: v(x, y=0.5)
FlowTrace
Ghia et al. 1982
0.2
0.1
0.0
−0.1
−0.2
v
0.0
0.2
0.4
0.6
0.8
1.0
x

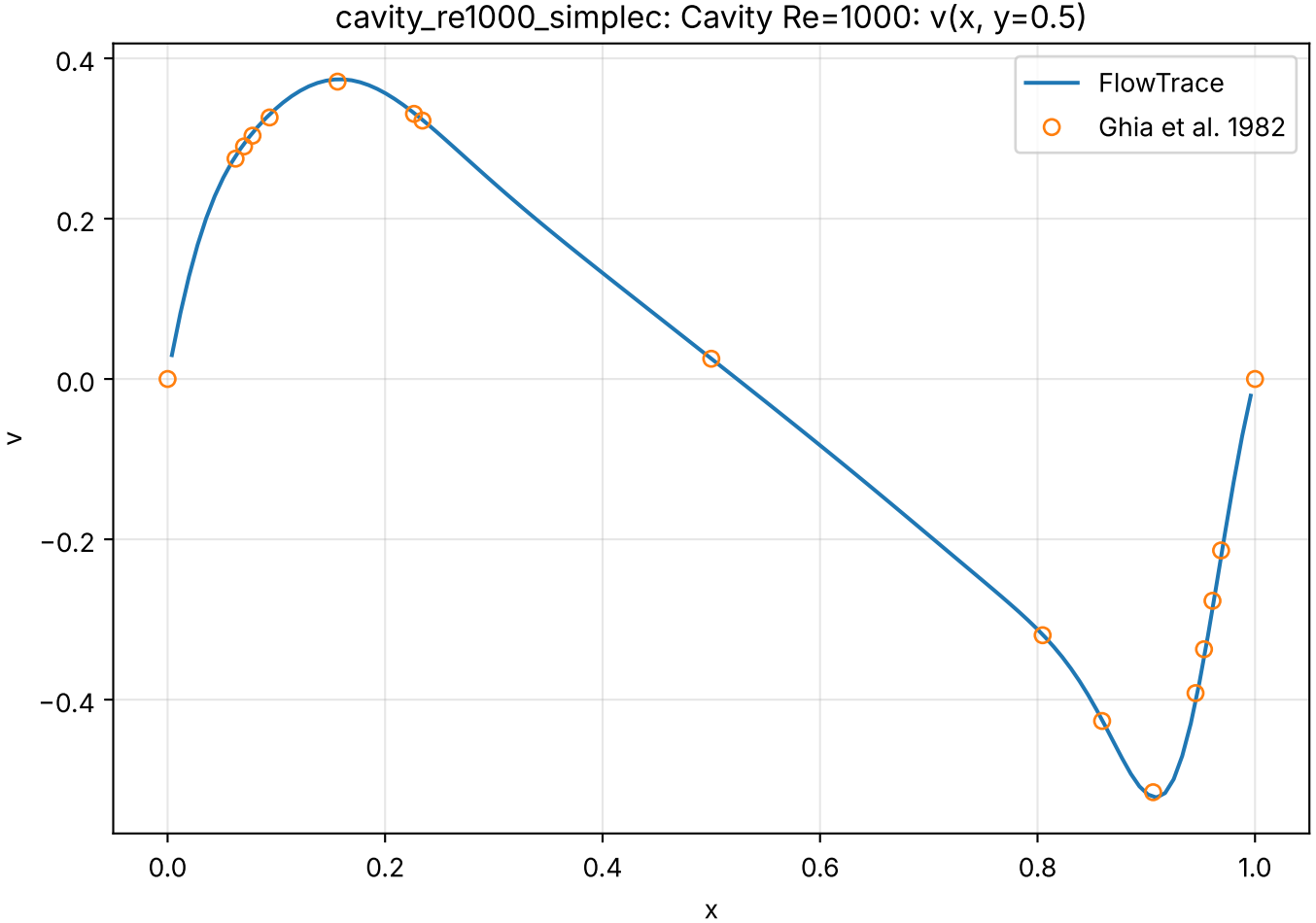

cavity_re100_simplec_quads: Cavity Re=100: u(x=0.5, y)
FlowTrace
Ghia et al. 1982
u
y
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
0.0
0.2
0.4
0.6
0.8
1.0

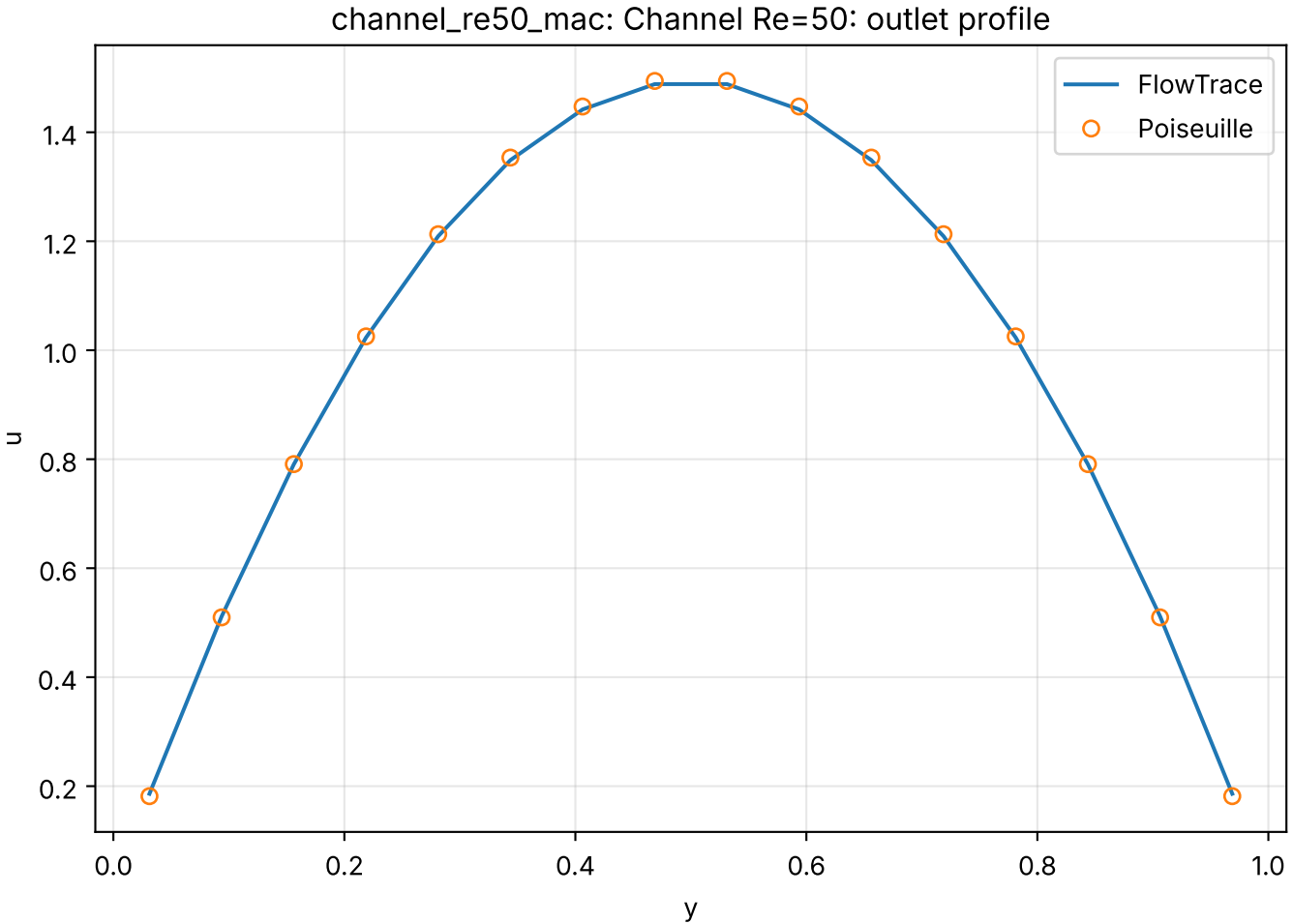

cavity_re100_simplec_quads: Cavity Re=100: v(x, y=0.5)
FlowTrace
Ghia et al. 1982
0.1
0.0
−0.1
−0.2
v
0.0
0.2
0.4
0.6
0.8
1.0
x

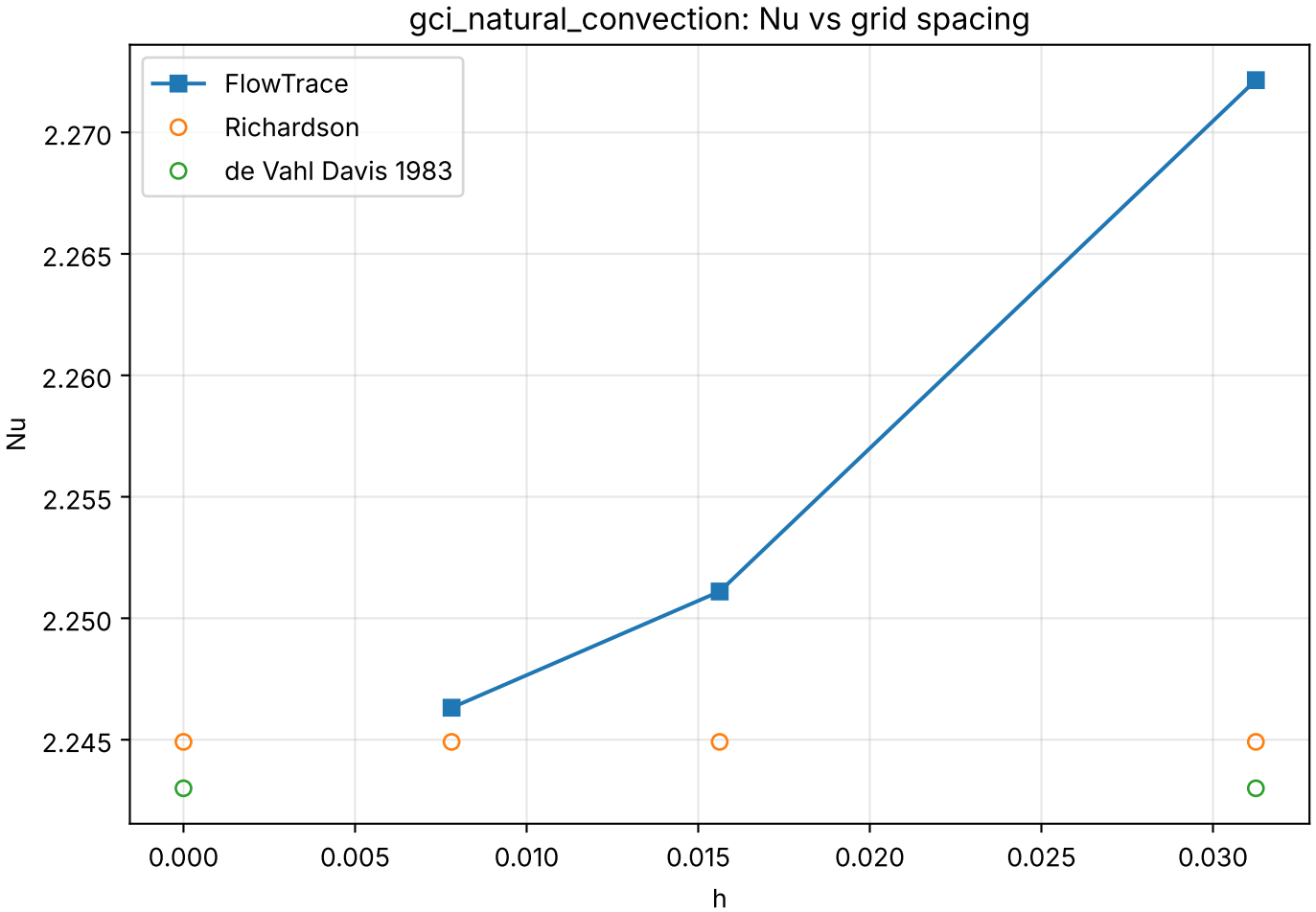

cavity_re100_simplec_tri: Cavity Re=100: u(x=0.5, y)
FlowTrace
Ghia et al. 1982
u
y
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
0.0
0.2
0.4
0.6
0.8
1.0

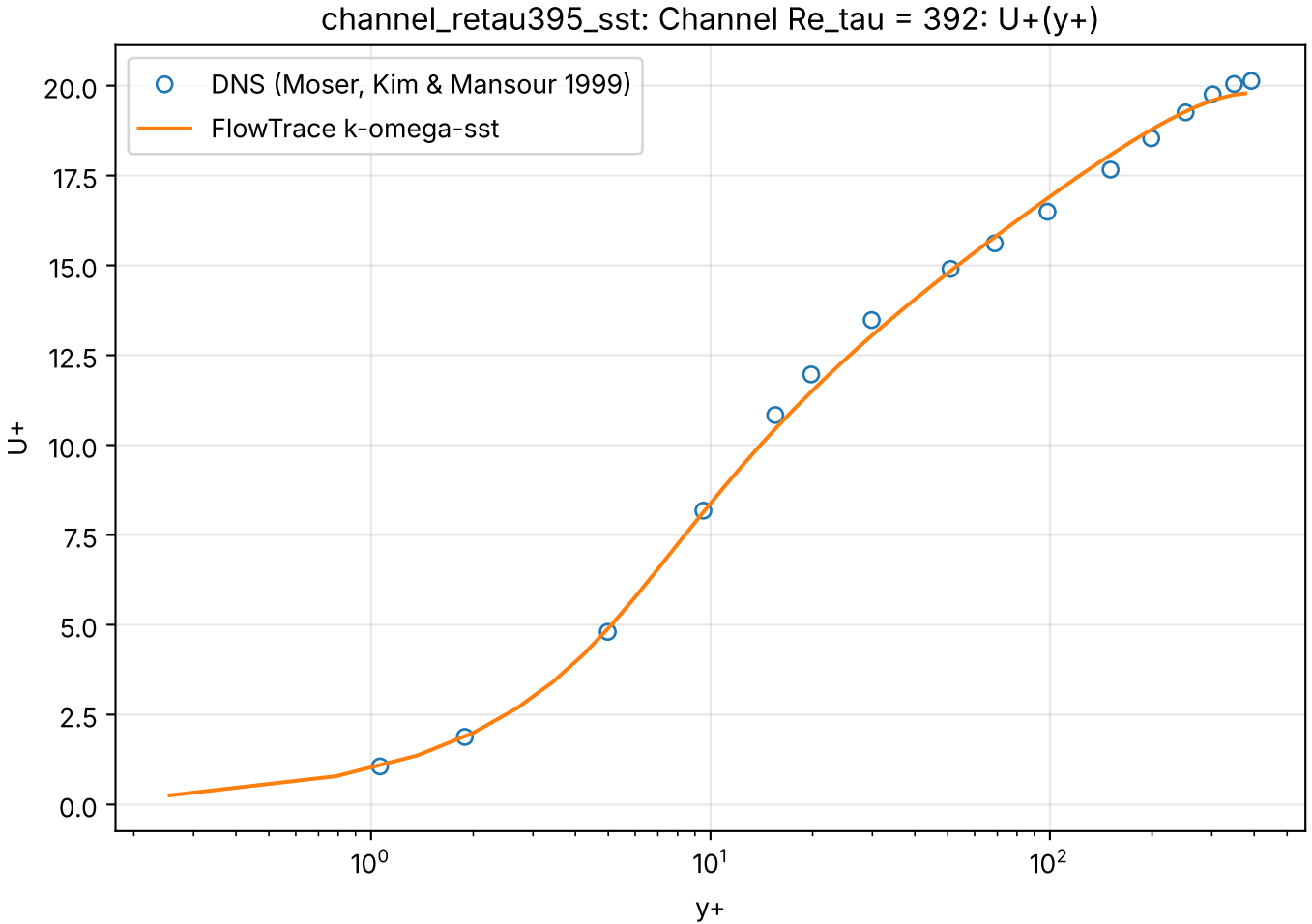

cavity_re100_simplec_tri: Cavity Re=100: v(x, y=0.5)
FlowTrace
Ghia et al. 1982
0.2
0.1
0.0
−0.1
−0.2
v
0.0
0.2
0.4
0.6
0.8
1.0
x

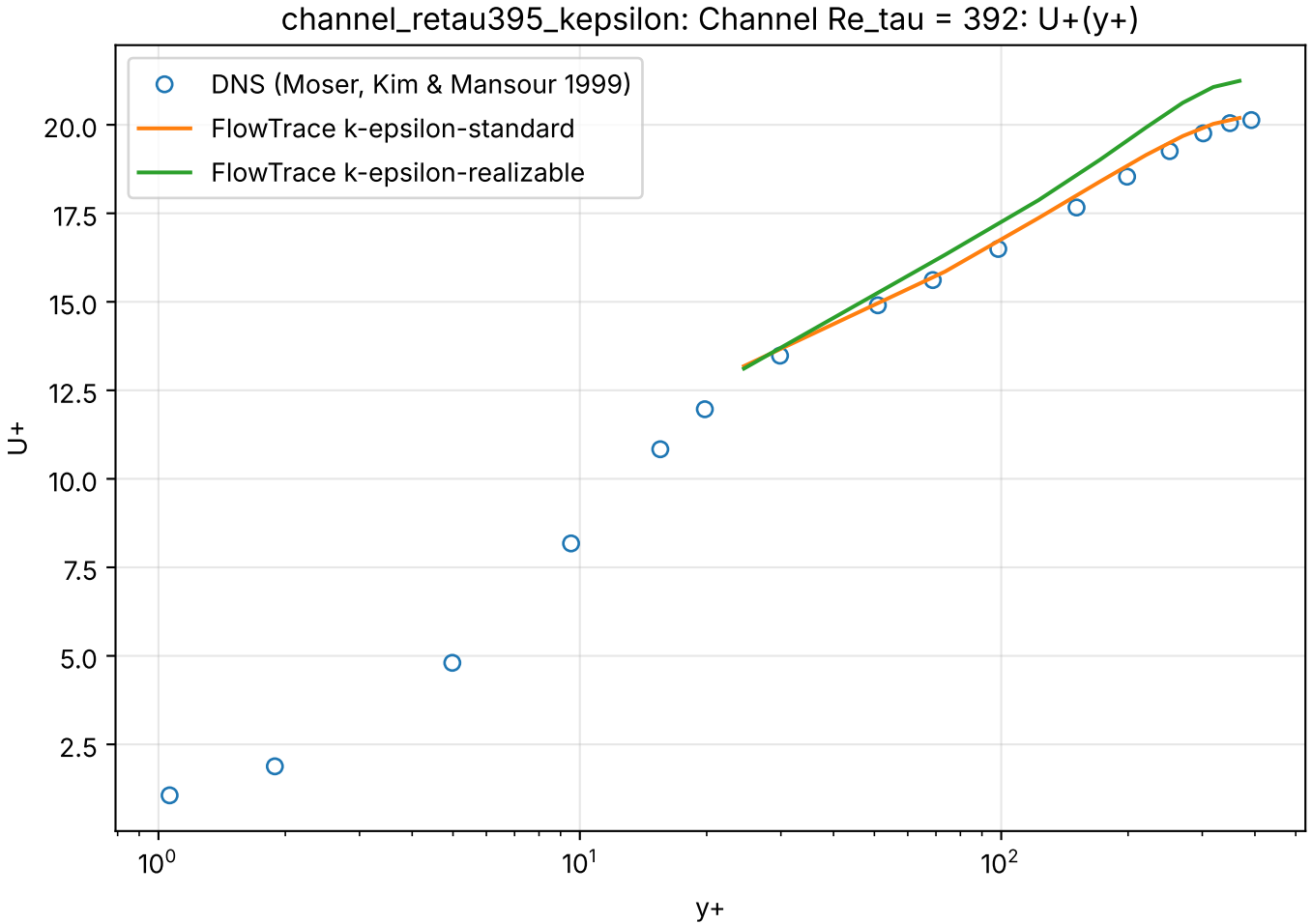

cavity_re1000_simplec: Cavity Re=1000: u(x=0.5, y)
FlowTrace
Ghia et al. 1982
u
y
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
0.0
0.2
0.4
0.6
0.8
1.0

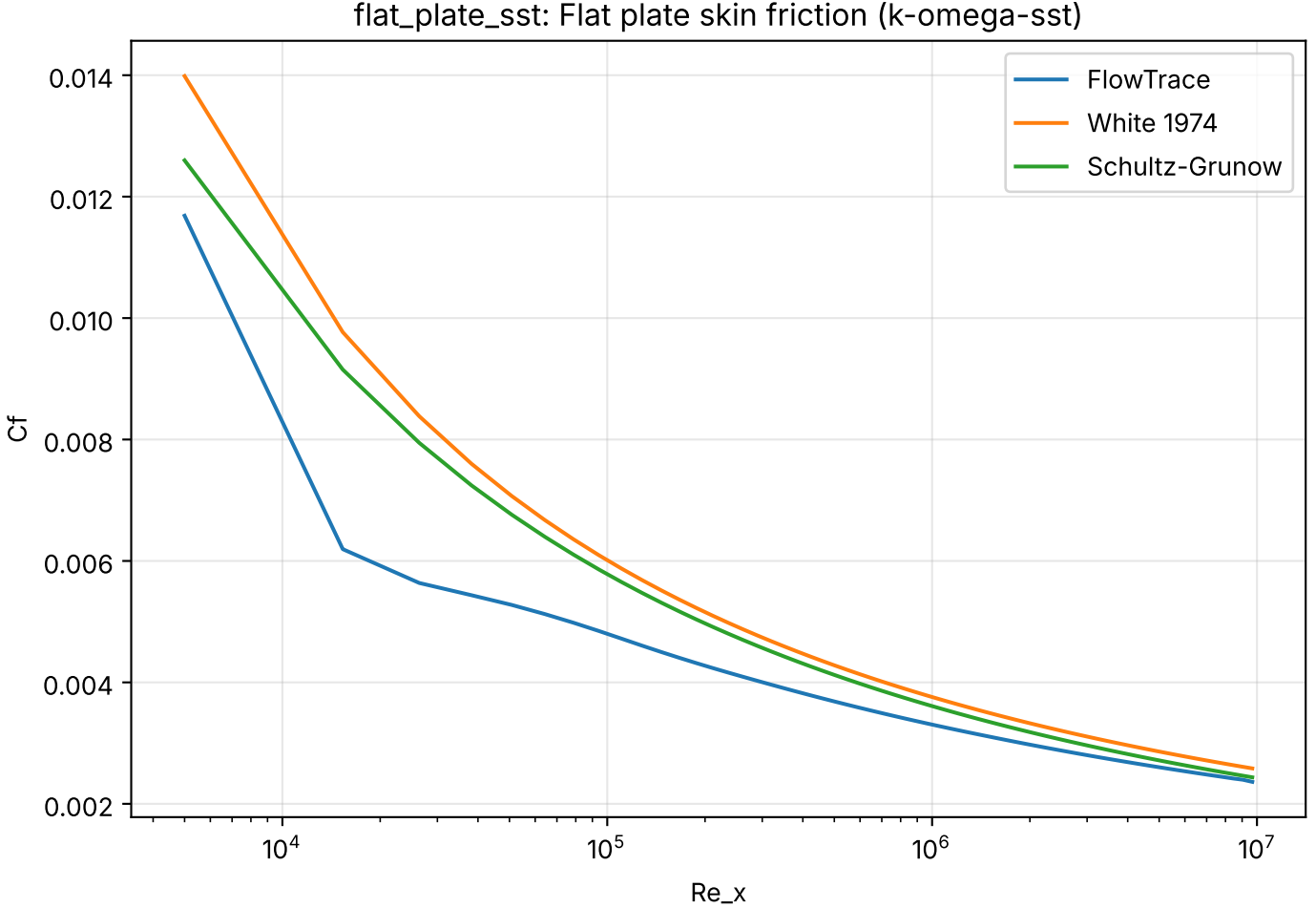

cavity_re1000_simplec: Cavity Re=1000: v(x, y=0.5)
FlowTrace
Ghia et al. 1982
0.4
0.2
0.0
−0.2
−0.4
v
0.0
0.2
0.4
0.6
0.8
1.0
x

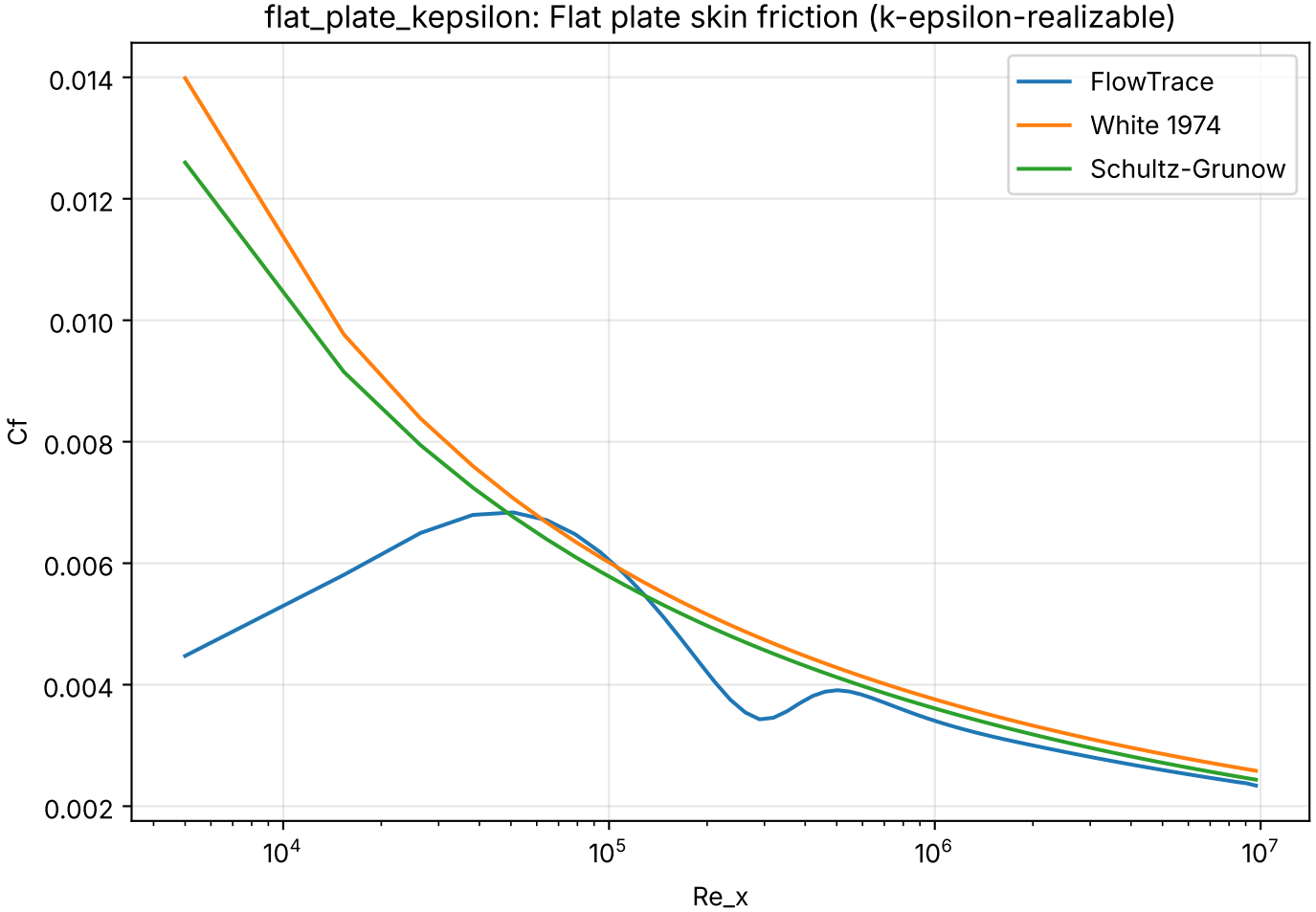

channel_re50_mac: Channel Re=50: outlet profile
FlowTrace
Poiseuille
u
y
0.2
0.4
0.6
0.8
1.0
1.2
1.4
0.0
0.2
0.4
0.6
0.8
1.0

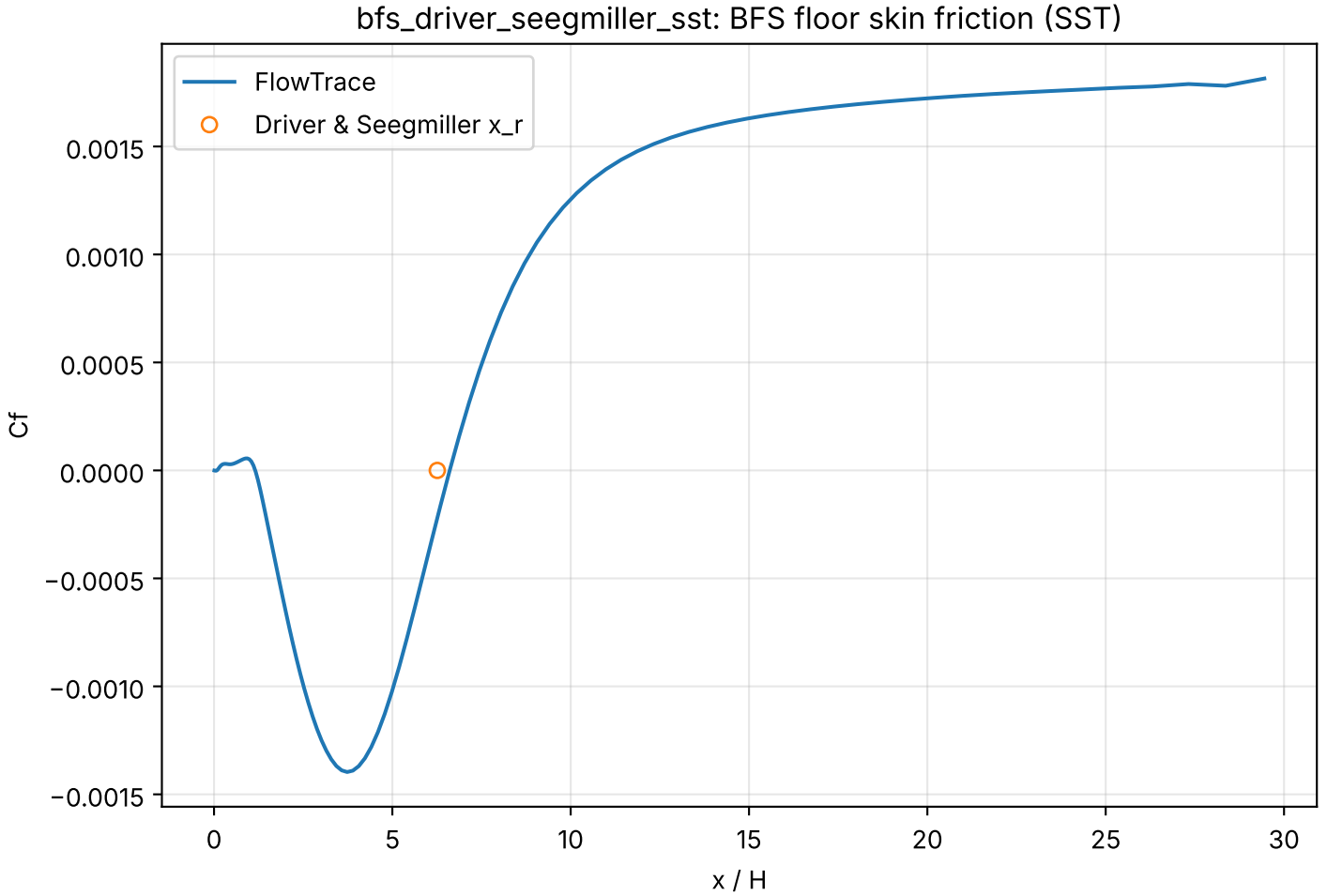

gci_natural_convection: Nu vs grid spacing
FlowTrace
Richardson
de Vahl Davis 1983
Nu
2.270
2.265
2.260
2.255
2.250
2.245
0.000
0.005
0.010
0.015
0.020
0.025
0.030
h


channel_retau395_sst: Channel Re_tau = 392: U+(y+)
DNS (Moser, Kim & Mansour 1999)
FlowTrace k-omega-sst
U+
y+
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
10^0
10^1
10^2


channel_retau395_kepsilon: Channel Re_tau = 392: U+(y+)
DNS (Moser, Kim & Mansour 1999)
FlowTrace k-epsilon-standard
FlowTrace k-epsilon-realizable
U+
y+
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
10^0
10^1
10^2


flat_plate_sst: Flat plate skin friction (k-omega-sst)
FlowTrace
White 1974
Schultz-Grunow
Cf
0.014
0.012
0.010
0.008
0.006
0.004
0.002
10^4
10^5
10^6
10^7
Re_x


flat_plate_kepsilon: Flat plate skin friction (k-epsilon-realizable)
FlowTrace
White 1974
Schultz-Grunow
0.014
0.012
0.010
0.008
0.006
0.004
0.002
Cf
10^4
10^5
10^6
10^7
Re_x


bfs_driver_seegmiller_sst: BFS floor skin friction (SST)
FlowTrace
Driver & Seegmiller x_r
0.0015
0.0010
0.0005
0.0000
−0.0005
−0.0010
−0.0015
Cf
0
5
10
15
20
25
30
x / H